STAT230A Lecture 7 Gauss-Markov Model (Parameter Estimation)

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1 Gauss-Markov Model 的 Parameter Estimation

Logic

接下来我们研究对 Gauss-Markov model 的 parameters β 和 σ^2 的 estimation

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Logic

我们已经知道 $\hat{\beta} = (X^\top X)^{-1} X^\top \vec{Y}$ 是 β 的 unbiased estimator, 我们希望进一步知道它是不是 β 最好的 unbiased estimator

1.1 Theorem: Gauss-Markov Theorem

对于 Gauss-Markov model, $\hat{\beta}$ 是 the **best linear unbiased estimator (BLUE)**, 即

对于任意满足以下条件的 $\tilde{\beta}$:

- **Linear:** $\tilde{\beta} = A\vec{Y}$ for some $A \in \mathbb{R}^{p \times n}$ not depending on $\vec{Y}$
- **Unbiased:** $\mathbb{E}[\tilde{\beta}] = \beta$

有:

$$\mathrm{Cov}(\hat{\beta}) \leq \mathrm{Cov}(\tilde{\beta}) \quad (\text{matrix comparison})$$

Remark: Some Implications

1. 若 $\mathrm{Cov}(\hat{\beta}) \leq \mathrm{Cov}(\tilde{\beta})$, 则 $\mathrm{Var}[\hat{\beta}_j] \leq \mathrm{Var}[\tilde{\beta}_j], \quad \forall j \in \{1, \ldots, p\}$

   Proof

$$\begin{aligned}
& \mathrm{Cov}(\hat{\beta}) \leq \mathrm{Cov}(\tilde{\beta}) \\
\Rightarrow \quad & \mathrm{Cov}(\tilde{\beta}) - \mathrm{Cov}(\hat{\beta}) \succeq 0 \\
\Rightarrow \quad & \vec{a}^\top (\mathrm{Cov}(\tilde{\beta}) - \mathrm{Cov}(\hat{\beta}))\vec{a} \geq 0 & \forall \vec{a} \in \mathbb{R}^p \\
\Rightarrow \quad & \vec{e}_j^\top (\mathrm{Cov}(\tilde{\beta}) - \mathrm{Cov}(\hat{\beta}))\vec{e}_j \geq 0 & \forall j \in \{1, \ldots, p\} \quad (\vec{e}_j = (0, \ldots, 0, 1, 0, \ldots, 0)^\top) \\
\Rightarrow \quad & \mathrm{Var}[\hat{\beta}_j] \leq \mathrm{Var}[\tilde{\beta}_j] & \forall j \in \{1, \ldots, p\}
\end{aligned}$$

2. Linear 是对于 Y 而言的 (不是对于 X), A 可以是 non-linear function of X, 例如 $A = (X^\top X)^{-1} X^\top$
3. $\tilde{\beta}$ 必须是 any true β 的 unbiased estimator, 因此无法选取 $A = 0$, 一个潜在的 $\tilde{\beta}$ 是 weighted linear square

Proof

对于任意 unbiased estimator $\tilde{\beta} = AY$, 有

$$\begin{aligned}
& \mathbb{E}[A\vec{Y}] = \beta & \forall \beta \\
\Rightarrow \quad & A\mathbb{E}[\vec{Y}] = \beta & \forall \beta \\
\Rightarrow \quad & AX\beta = \beta & \forall \beta \\
\Rightarrow \quad & AX = I
\end{aligned}$$

接下来证明 $\mathrm{Cov}(\tilde{\beta}) - \mathrm{Cov}(\hat{\beta}) \succeq 0$, 注意到:

$$
\begin{aligned}
\mathrm{Cov}(\tilde{\beta})-\mathrm{Cov}(\hat{\beta}) &= \mathrm{Cov}(\hat{\beta}+\tilde{\beta}-\hat{\beta})-\mathrm{Cov}(\hat{\beta}) \\
&= \mathrm{Cov}(\hat{\beta})+\mathrm{Cov}(\tilde{\beta}-\hat{\beta},\hat{\beta})+\mathrm{Cov}(\hat{\beta},\tilde{\beta}-\hat{\beta})+\mathrm{Cov}(\tilde{\beta}-\hat{\beta})-\mathrm{Cov}(\hat{\beta}) \\
&= \mathrm{Cov}(\tilde{\beta}-\hat{\beta},\hat{\beta})+\mathrm{Cov}(\hat{\beta},\tilde{\beta}-\hat{\beta})+\mathrm{Cov}(\tilde{\beta}-\hat{\beta})
\end{aligned}
$$

由于对任意 random vector $\vec{v}$, 有 $\mathrm{Cov}(\vec{v},\vec{v})\succeq 0$, 因此:

$$
\mathrm{Cov}(\tilde{\beta}-\hat{\beta})\succeq 0
$$

而对于 $\mathrm{Cov}(\hat{\beta},\tilde{\beta}-\hat{\beta})$, 记 $\hat{A}_{OLS}:=(X^\top X)^{-1}X^\top$, 有:

$$
\begin{aligned}
\mathrm{Cov}(\hat{\beta},\tilde{\beta}-\hat{\beta}) &= \mathrm{Cov}(\hat{A}_{OLS}Y, AY-\hat{A}_{OLS}Y) \\
&= \hat{A}_{OLS}\mathrm{Cov}(Y)(A-\hat{A}_{OLS})^\top \\
&= \sigma^2\hat{A}_{OLS}(A-\hat{A}_{OLS})^\top \\
&= \sigma^2(\hat{A}_{OLS}A^\top-\hat{A}_{OLS}\hat{A}_{OLS}^\top) \\
&= \sigma^2((X^\top X)^{-1}X^\top A^\top-(X^\top X)^{-1}X^\top X(X^\top X)^{-1}) \\
&= \sigma^2((X^\top X)^{-1}(X^\top A^\top-I)) \\
&= 0 && (\text{由于 } AX=I)
\end{aligned}
$$

类似地,

$$
\mathrm{Cov}(\tilde{\beta}-\hat{\beta},\hat{\beta})=0
$$

因此,

$$
\begin{aligned}
& \mathrm{Cov}(\tilde{\beta})-\mathrm{Cov}(\hat{\beta})=\mathrm{Cov}(\tilde{\beta}-\hat{\beta})\succeq 0 \\
\Rightarrow\quad & \mathrm{Cov}(\hat{\beta})\preceq\mathrm{Cov}(\tilde{\beta})
\end{aligned}
$$

Logic

我们已经知道 $\hat{\beta}$ 是 β 的 best linear unbiased estimator, 接下来我们研究 σ^2 的 unbiased estimator

1.2 σ^2 的 Unbiased Estimator

对于 error variance σ^2, 其 unbiased estimator 为:

$$
\hat{\sigma}^2=\frac{\sum_{i=1}^n\hat{\varepsilon}_i^2}{n-p}=\frac{\vec{\hat{\varepsilon}}^\top\vec{\hat{\varepsilon}}}{n-p}
$$

Proof

考虑 sum of $\hat{\varepsilon}_i^2$,

$$
\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^n\hat{\varepsilon}_i^2\right] &= \sum_{i=1}^n\mathbb{E}[\hat{\varepsilon}_i^2] \\
&= \sum_{i=1}^n\mathrm{Var}[\hat{\varepsilon}_i] && (\mathbb{E}[\hat{\varepsilon}_i]=0) \\
&= \sum_{i=1}^n\sigma^2(1-h_{ii}) \\
&= \sigma^2\left(n-\sum_{i=1}^n h_{ii}\right) \\
&= \sigma^2(n-\mathrm{Tr}(H)) \\
&= \sigma^2(n-p) && (\text{若 } X^\top X \text{ 为 full rank})
\end{aligned}
$$

因此,

$$
\mathbb{E}[\hat{\sigma}^2]=\mathbb{E}\left[\frac{\sum_{i=1}^n\hat{\varepsilon}_i^2}{n-p}\right]=\sigma^2
$$